

Simplifying Surds

AQA · KS4 Year 10–11

Name: _____ Class: _____ Date: _____

What you need to be able to do:

Simplify surds by finding the largest square factor. Express surds in their simplest form $a\sqrt{b}$. Understand why surds are 'exact' values and when to leave answers in surd form.

Common Mistakes to Avoid:

- Not finding the LARGEST square factor — $72 = (36 \times 2) = 6\sqrt{2}$, not $(4 \times 18) = 2\sqrt{18}$ (not fully simplified).
- Thinking $(a + b) = a + b$ — this is WRONG. $(9 + 16) \neq 9 + 16$.
- Writing $4 \times 3 = 12$ is correct, but the point of simplifying is to go the OTHER way: $12 = 4 \times 3 = 2\sqrt{3}$.
- Forgetting that $a \times a = a^2$ (not a^2).

Method Steps:

1. Step 1: Find the largest perfect square that is a factor of the number under the $\sqrt{\quad}$.
2. Step 2: Split: $n = (\text{square factor} \times \text{remainder})$.
3. Step 3: Use $(a \times b) = a \times b$.
4. Step 4: Take the square root of the perfect square factor out.
5. Step 5: Write answer as $k\sqrt{m}$ where m has no square factors.

Worked Example:

Simplify $\sqrt{72}$.

Step 1 — Largest square factor of 72: 36 (since $36 \times 2 = 72$).

Step 2 — $72 = (36 \times 2)$

Step 3 — $= \sqrt{36 \times 2}$

Step 4 — $= 6\sqrt{2}$

Check: $6^2 \times 2 = 36 \times 2 = 72$

Question 1 [2 marks]

Simplify:

- a) 12 b) 18 c) 20 d) 50

Answer lines

Question 2 [2 marks]

Simplify:

- a) 8 b) 32 c) 45 d) 75

Answer lines

Question 3 [3 marks]

Simplify:

- a) 98 b) 200 c) 128 d) 300

Answer lines

Question 4 [3 marks]

Simplify:

- a) (4×7) b) $3 \cdot 12$ c) $2 \cdot 50$ d) $5 \cdot 8$

Answer lines

Question 5 [3 marks]

Simplify fully:

a) $48 + 12$ b) $75 - 27$ c) $2 \times 18 + 3 \times 2$

Answer lines

Question 6 [4 marks]

Show that:

a) $80 = 4 \cdot 5$ b) $3 \cdot 28 = 6 \cdot 7$ c) $180 = 6 \cdot 5$

Answer lines

Question 7 [4 marks]

A square has area 72 cm^2 .

- a) Find the exact side length. (1 mark)
- b) Find the exact perimeter. (1 mark)
- c) A second square has side length $3\sqrt{2} \text{ cm}$. Which square is larger? Show working. (2 marks)

Answer lines

Question 8 [5 marks]

a) Show that $(a^2b)^3 = a^6b^3$ for any positive a and b . (1 mark)

b) Simplify $(50x^4y^2)^{1/2}$ where $x, y > 0$. (2 marks)

c) A right-angled triangle has shorter sides 18 cm and 32 cm. Find the exact hypotenuse in simplest surd form. (2 marks)

Answer lines

Self-reflection

How confident do you feel with this topic now? (tick one)

Not yet Getting there Confident Really confident

Which question did you find the hardest, and why?

Do I always find the LARGEST square factor first time?

One thing I will practise before the next lesson:

Teacher key

Q1: a) 2 3 b) 3 2 c) 2 5 d) 5 2

Q2: a) 2 2 b) 4 2 c) 3 5 d) 5 3

Q3: a) 7 2 b) 10 2 c) 8 2 d) 10 3

Q4: a) 2 7 b) $3 \times 2 = 6$ 3 c) $2 \times 5 = 10$ 2 d) $5 \times 2 = 10$ 2

Q5: a) 4 3 + 2 3 = 6 3 b) 5 3 - 3 3 = 2 3 c) 6 2 + 3 2 = 9 2

Q6: a) $80 = (16 \times 5) = 4 \times 5$ b) $3 \times 28 = 3 \times (4 \times 7) = 3 \times 28 = 84$ c) $180 = (36 \times 5) = 6 \times 5$

Q7: a) $72 = 6^2$ cm b) $4 \times 6 = 24$ cm c) Second square side = 3, first = 6. $6^2 > 3^2$, so first is larger.

Q8: a) $(a^2b) = (a^2) \times b = a^2b$ b) $(50x^4y^2) = (25 \times 2 \times x^4 \times y^2) = 5x^2y^2$ c) Hyp² = $18 + 32 = 50$; hyp = $\sqrt{50} = 5\sqrt{2}$ cm